

PROJECT 3 SERVER AND SERVICES

GURUCHANDHRAN M

TASK 1: CREATING STATIC ROUTES IN THE ROUTER

ROUTER 1

R1 IP: 10.0.0.10 - eth0/0

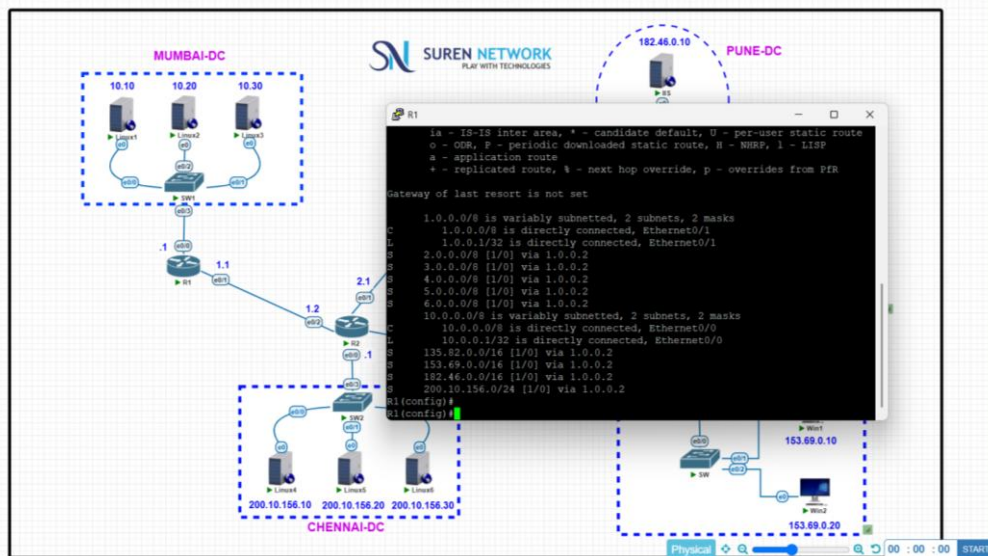
STATIC ROUTE R1:

2.0.0.0/8	1.0.0.2
3.0.0.0/8	1.0.0.2
4.0.0.0/8	1.0.0.2
5.0.0.0/8	1.0.0.2
6.0.0.0/8	1.0.0.2

R1 IP: 1.0.0.1 - eth0/1

STATIC ROUTE R1:

135.82.0.0/16	1.0.0.2
153.69.0.0/16	1.0.0.2
182.46.0.0/16	1.0.0.2
200.10.156.0/24	1.0.0.2



ROUTER 2

R2 IP: 200.10.156.1 - eth0/0

R2 IP: 1.0.0.2 - eth0/2

STATIC ROUTE R2:

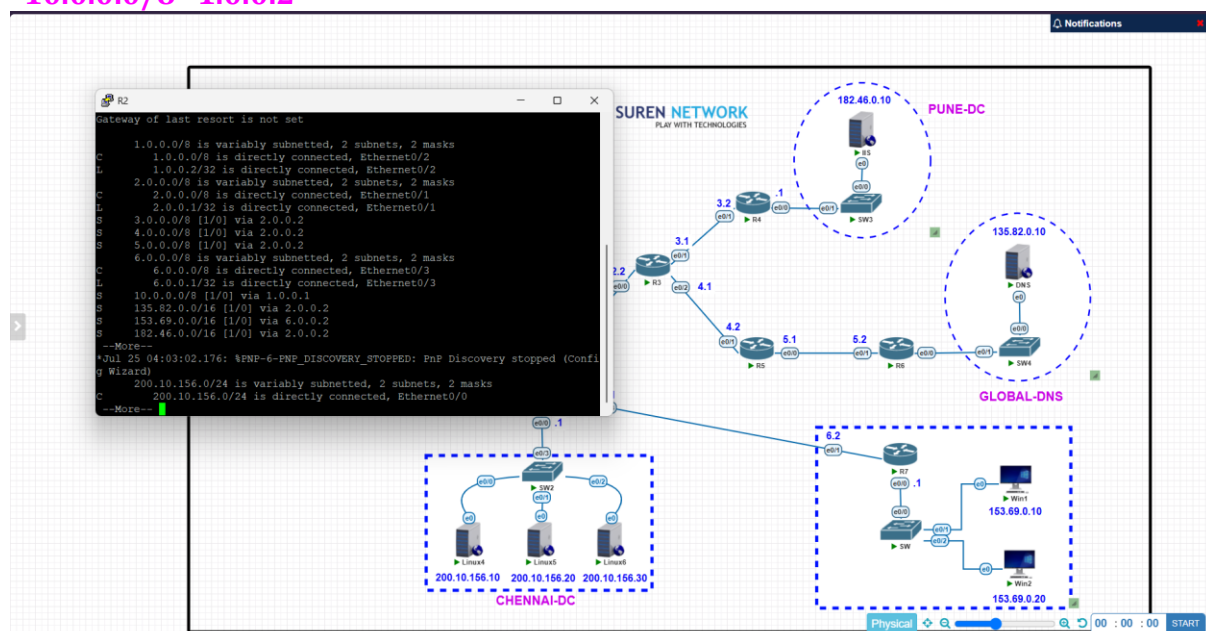
3.0.0.0/8	1.0.0.2
4.0.0.0/8	1.0.0.2
5.0.0.0/8	1.0.0.2
10.0.0.0/8	1.0.0.2

R2 IP: 2.0.0.1 - eth0/1

R2 IP: 6.0.0.1 - eth0/3

STATIC ROUTE R2:

135.82.0.0/16	2.0.0.2
153.69.0.0/16	6.0.0.2
182.46.0.0/16	2.0.0.2



ROUTER 3

R3 IP: 2.0.0.2 – eth0/0

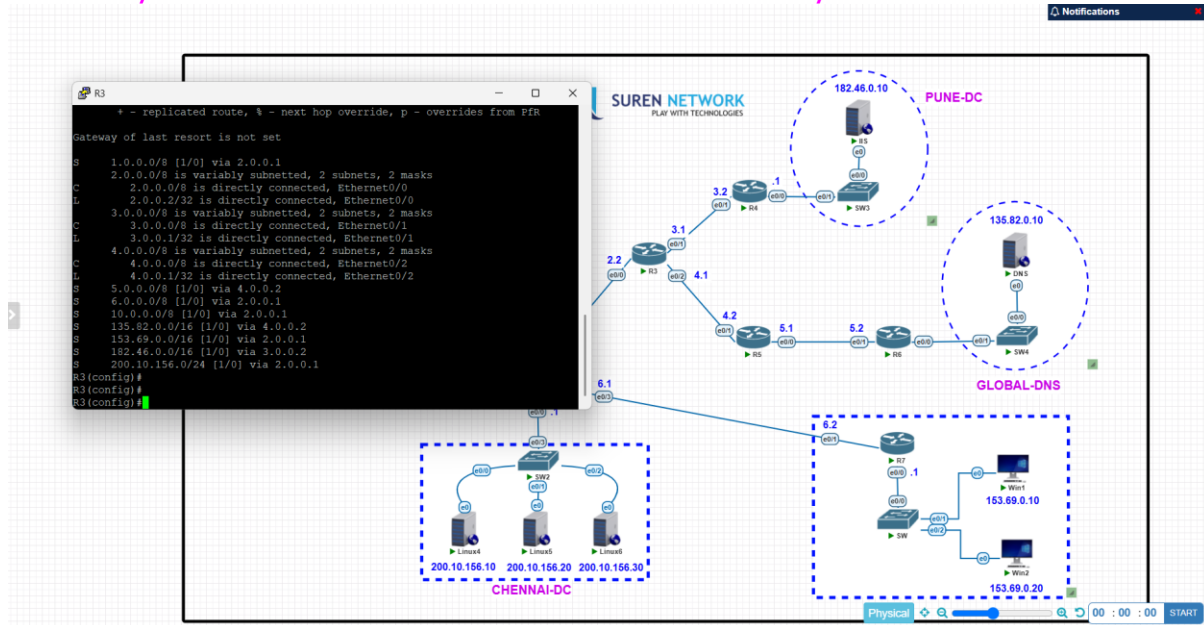
STATIC ROUTE R3:

1.0.0.0/8	2.0.0.1
5.0.0.0/8	4.0.0.2
6.0.0.0/8	2.0.0.1
10.0.0.0/8	2.0.0.1

R3 IP: 3.0.0.1 – eth0/1

STATIC ROUTE R3:

135.82.0.0/16	4.0.0.2
153.69.0.0/16	2.0.0.1
182.46.0.0/16	3.0.0.2
200.10.156.0/24	2.0.0.1



ROUTER 4

R4 IP: 182.46.0.1 – eth0/0

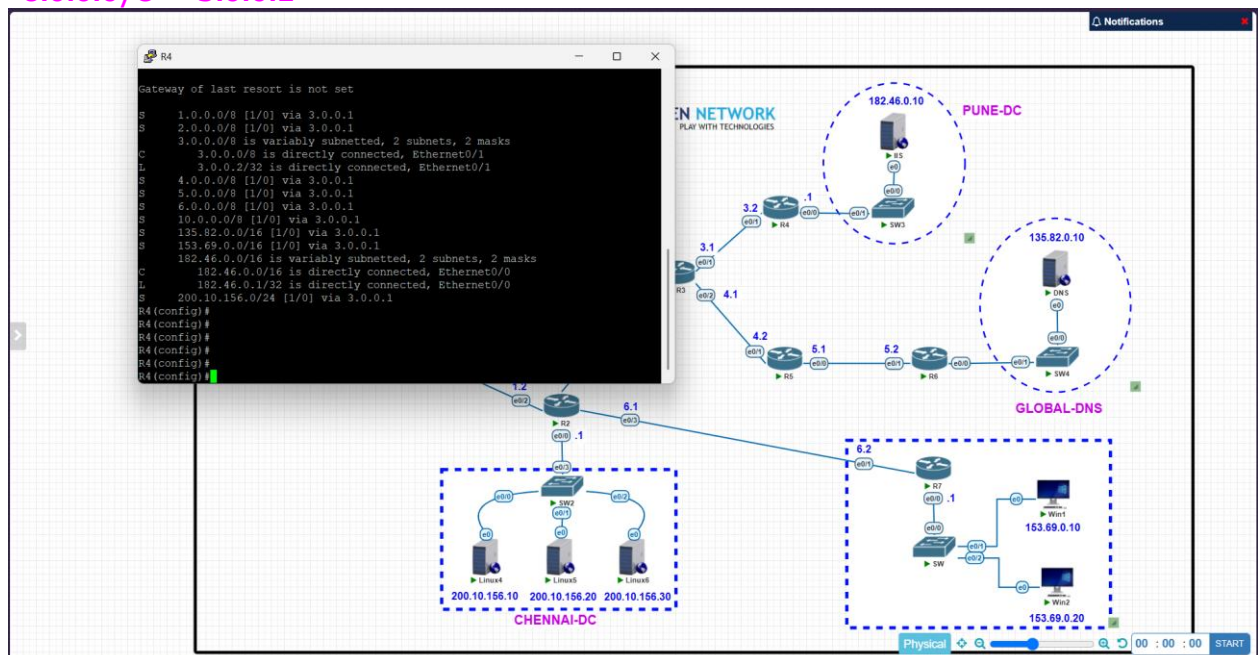
STATIC ROUTE R4:

1.0.0.0/8	3.0.0.1
2.0.0.0/8	3.0.0.1
4.0.0.0/8	3.0.0.1
5.0.0.0/8	3.0.0.1
6.0.0.0/8	3.0.0.1

R4 IP: 3.0.0.2 – eth0/1

STATIC ROUTE R4:

10.0.0.0/8	3.0.0.1
135.82.0.0/16	3.0.0.1
153.69.0.0/16	3.0.0.1
200.10.156.0/24	3.0.0.1



ROUTER 5

R5 IP: 4.0.0.1 – eth0/0

STATIC ROUTE R5:

1.0.0.0/8 4.0.0.1

2.0.0.0/8 4.0.0.1

3.0.0.0/8 4.0.0.1

6.0.0.0/8 4.0.0.1

10.0.0.0/8 4.0.0.1

R5 IP: 5.0.0.2 – eth0/2

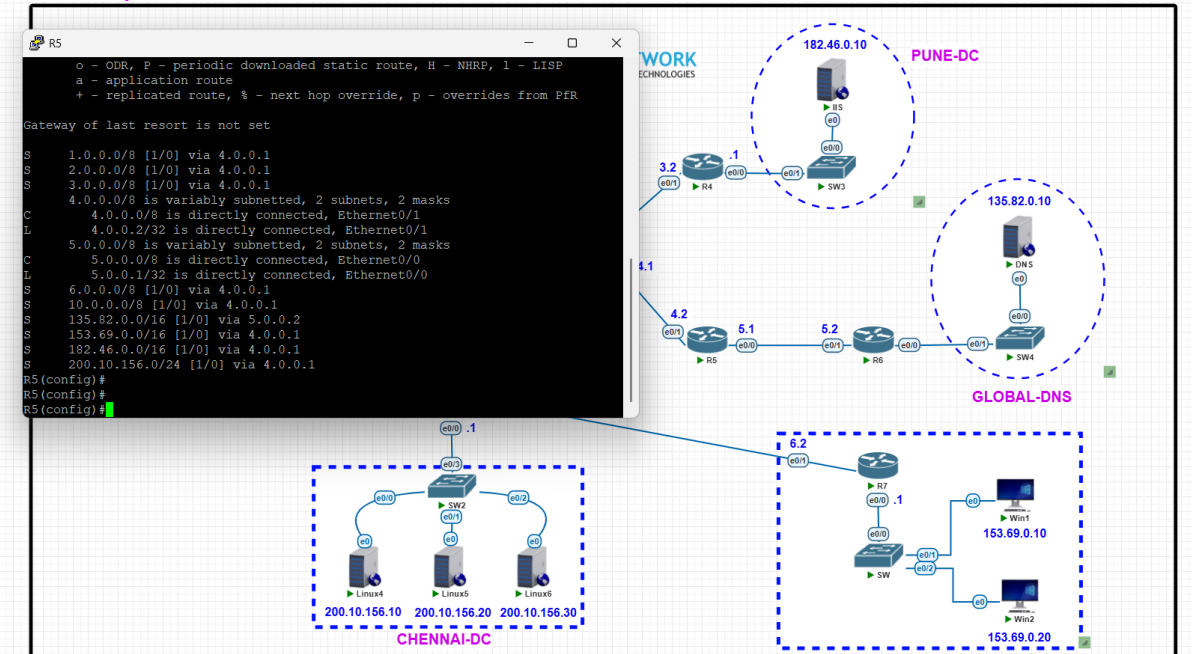
STATIC ROUTE R5:

135.82.0.0/16 5.0.0.2

153.69.0.0/16 4.0.0.1

182.46.0.0/16 4.0.0.1

200.10.156.0/24 4.0.0.1



ROUTER 6

R6 IP: 135.82.0.1 – eth0/0

STATIC ROUTE R6:

1.0.0.0/8 5.0.0.1

2.0.0.0/8 5.0.0.1

3.0.0.0/8 5.0.0.1

4.0.0.0/8 5.0.0.1

6.0.0.0/8 5.0.0.1

R6 IP: 5.0.0.2 – eth0/1

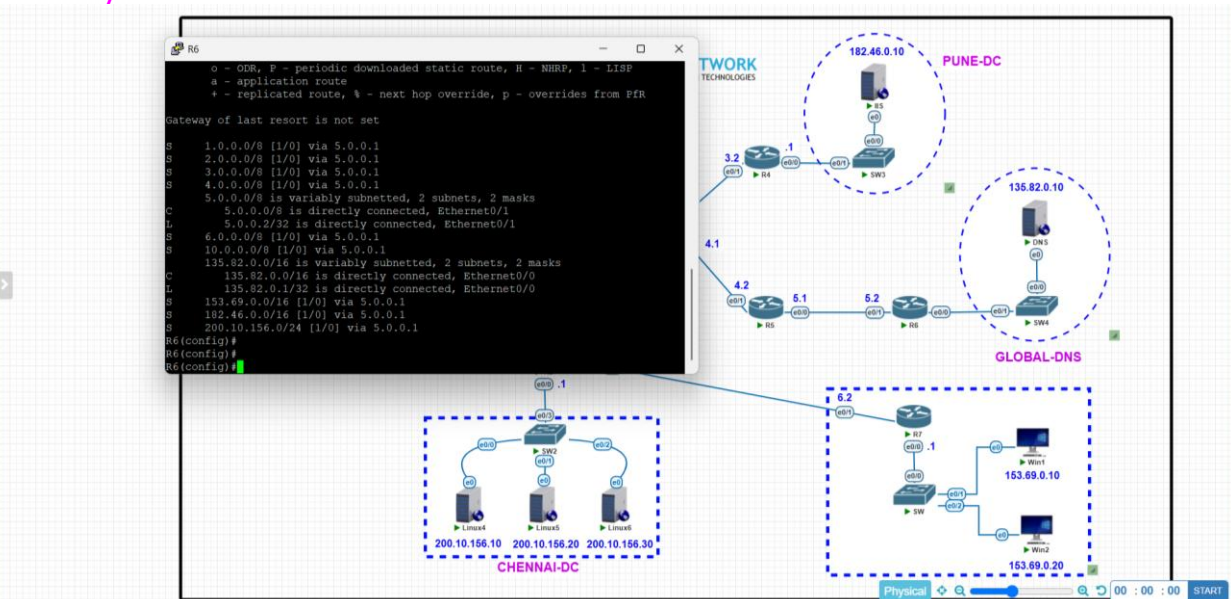
STATIC ROUTE R6:

10.0.0.0/8 5.0.0.1

153.69.0.0/16 5.0.0.1

182.46.0.0/16 5.0.0.1

200.10.156.0/24 5.0.0.1



ROUTER 7

R7 IP: 153.69.0.1 – eth0/0

STATIC ROUTE R7:

1.0.0.0/8 6.0.0.1

2.0.0.0/8 6.0.0.1

3.0.0.0/8 6.0.0.1

4.0.0.0/8 6.0.0.1

5.0.0.0/8 6.0.0.1

R7 IP: 6.0.0.2 – eth0/1

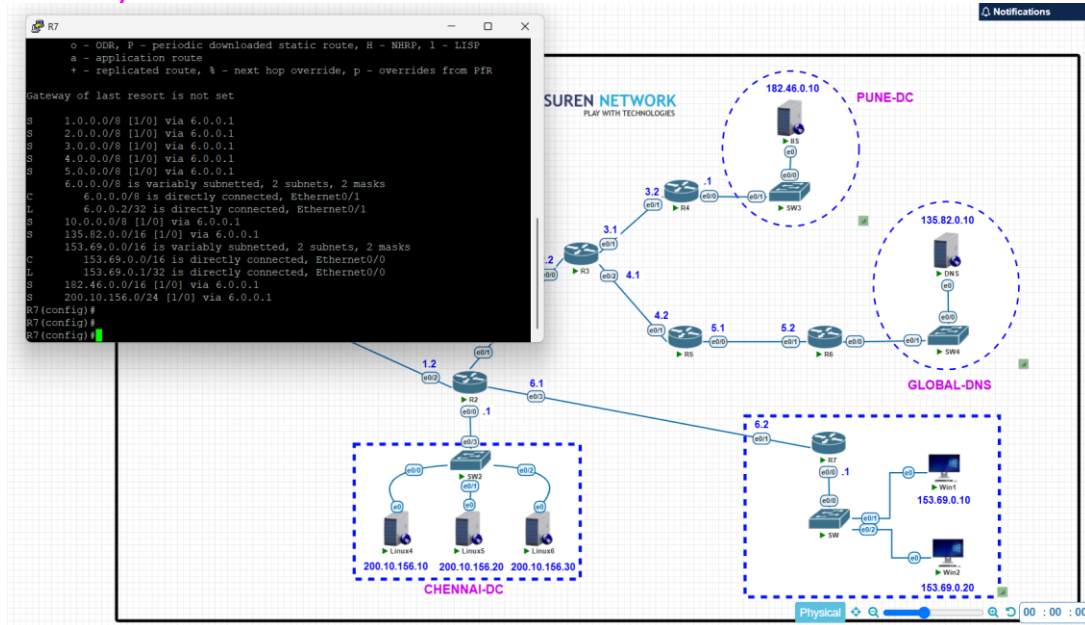
STATIC ROUTE R7:

10.0.0.0/8 6.0.0.1

135.82.0.0/16 6.0.0.1

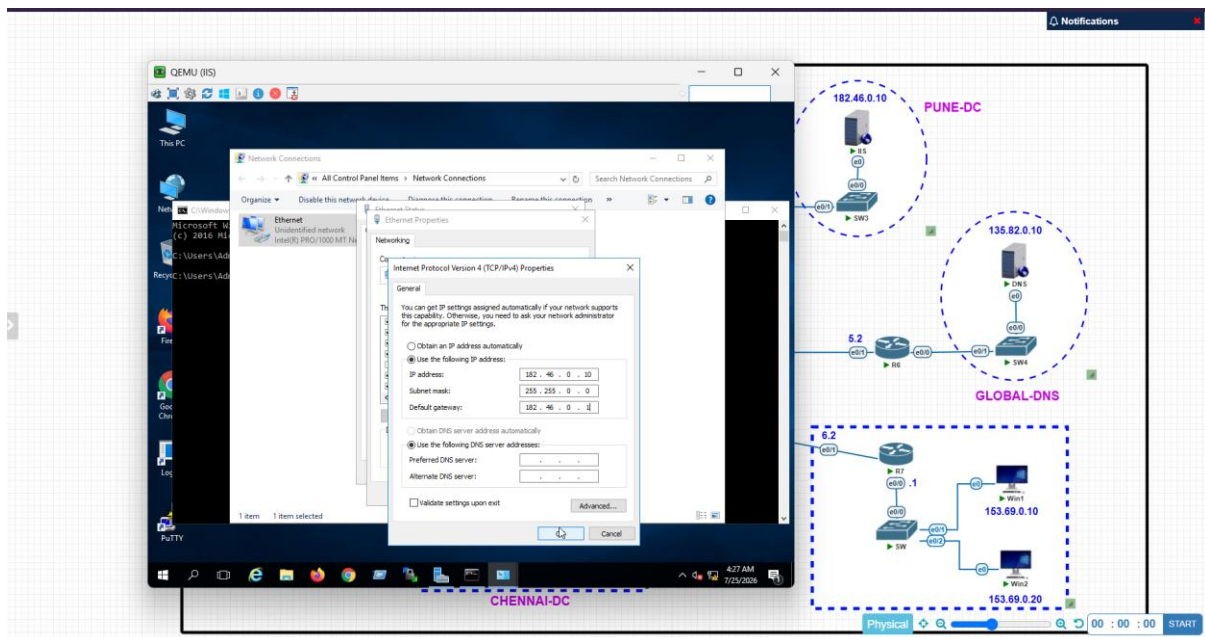
182.46.0.0/16 6.0.0.1

200.10.156.0/24 6.0.0.1

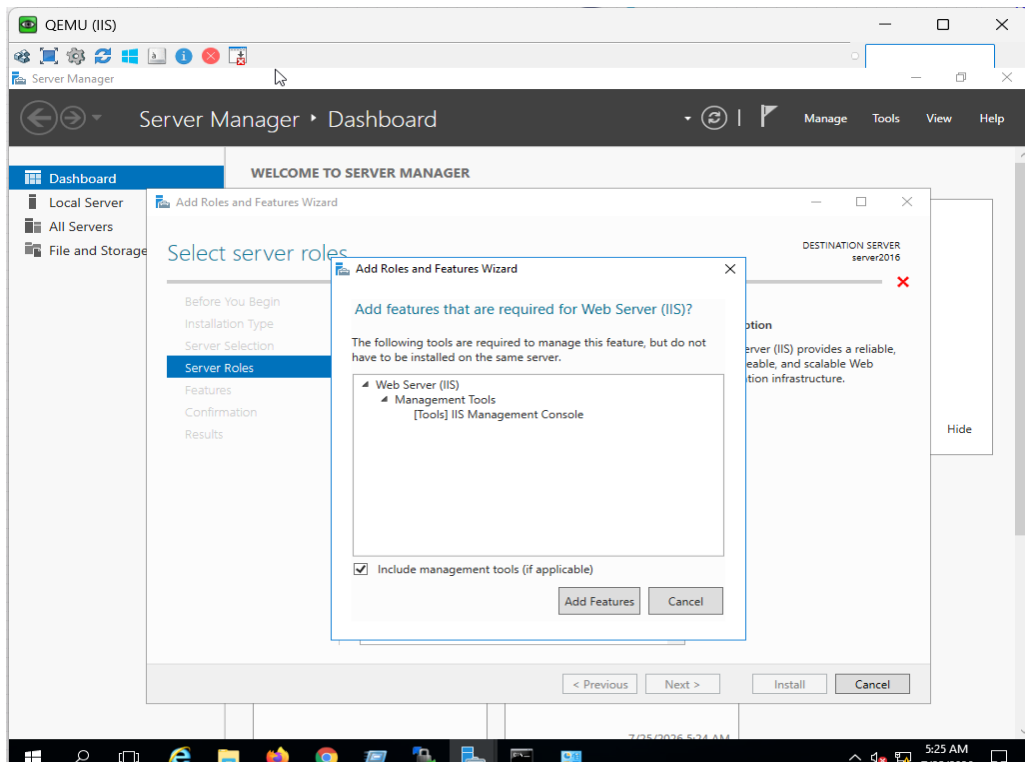


TASK 2: CONFIGURING THE WINDOWS (IIS) SERVER

IP ASSIGNMENT



INSTALLING IIS SERVICE



TASK 3: CONFIGURING THE LINUX SERVERS

LINUX SERVER 1

IP: 10.0.0.10

```
QEMU (Linux1)
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]

ifconfig eth0
Link encap:Ethernet HWaddr 00:50:00:00:01:00
inet addr:10.0.0.10 Bcast:10.255.255.255 Mask:255.0.0.0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:440 errors:0 dropped:0 overruns:0 frame:0
TX packets:223 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:141156 (137.0 KiB) TX bytes:71806 (70.1 KiB)

lo
Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)
```

LINUX SERVER 2

IP: 10.0.0.20

```
QEMU (Linux2)

ifconfig eth0
Link encap:Ethernet HWaddr 00:50:00:00:02:00
inet addr:10.0.0.20 Bcast:10.255.255.255 Mask:255.0.0.0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:589 errors:0 dropped:0 overruns:0 frame:0
TX packets:294 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:189134 (184.7 KiB) TX bytes:94668 (92.4 KiB)

lo
Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)
```

LINUX SERVER 3

IP: 10.0.0.30

```
QEMU (Linux3)
-bash: ifconfig: command not found
root@linux3:~# ifconfig
-bash: ifconfig: command not found
root@linux3:~# ifconfig
eth0
Link encap:Ethernet HWaddr 00:50:00:00:06:00
inet addr:10.0.0.30 Bcast:10.255.255.255 Mask:255.0.0.0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:615 errors:0 dropped:0 overruns:0 frame:0
TX packets:307 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:197506 (192.8 KiB) TX bytes:98854 (96.5 KiB)

lo
Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)
```

LINUX SERVER 4

IP: 200.10.156.10

```
QEMU (Linux4)

ifconfig eth0
Link encap:Ethernet HWaddr 00:50:00:00:03:00
inet addr:200.10.156.10 Bcast:200.10.156.255 Mask:255.255.255.0
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:650 errors:0 dropped:0 overruns:0 frame:0
TX packets:330 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:209300 (204.3 KiB) TX bytes:106260 (103.7 KiB)

lo
Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)
```

LINUX SERVER 5

IP: 200.10.156.20

```
QEMU (Linux5)
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]

ifconfig
eth0  Link encap:Ethernet  HWaddr 00:50:00:00:04:00
      inet addr:200.10.156.20 Bcast:200.10.156.255 Mask:255.255.255.0
      UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
      RX packets:897 errors:0 dropped:0 overruns:0 frame:0
      TX packets:447 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:1000
      RX bytes:286476 (279.7 KiB)  TX bytes:143934 (140.5 KiB)

lo    Link encap:Local Loopback
      inet addr:127.0.0.1 Mask:255.0.0.0
      UP LOOPBACK RUNNING  MTU:16436  Metric:1
      RX packets:0 errors:0 dropped:0 overruns:0 frame:0
      TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:0
      RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

LINUX SERVER 6

IP: 200.10.156.30

```
QEMU (Linux6)
shutting down Ethernet interface eth0 ..... [ OK ]
shutting down loopback interface lo ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
bringing up Ethernet interface eth0 ..... [ OK ]
set up default gateway ..... [ OK ]

ifconfig
eth0  Link encap:Ethernet  HWaddr 00:50:00:00:05:00
      inet addr:200.10.156.30 Bcast:200.10.156.255 Mask:255.255.255.0
      UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
      RX packets:905 errors:0 dropped:0 overruns:0 frame:0
      TX packets:451 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:1000
      RX bytes:289052 (282.2 KiB)  TX bytes:145222 (141.8 KiB)

lo    Link encap:Local Loopback
      inet addr:127.0.0.1 Mask:255.0.0.0
      UP LOOPBACK RUNNING  MTU:16436  Metric:1
      RX packets:0 errors:0 dropped:0 overruns:0 frame:0
      TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
      collisions:0 txqueuelen:0
      RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

UPDATING THE CONTENTS OF LINUX WEB SERVICE

LINUX SERVER 1

```
QEMU (Linux1)
Hi welcome to learn Linux.
This is the Linux Server 1
```

LINUX SERVER 2

```
QEMU (Linux2)
Hi welcome to learn Linux.
This is the Linux Server 2
```

LINUX SERVER 3

```
QEMU (Linux3)
Hi welcome to learn Linux.
This is the Linux Server 3
```

LINUX SERVER 4

```
QEMU (Linux4)
Hi welcome to learn Linux.
This is the Linux Server 4
```

LINUX SERVER 5

```
QEMU (Linux5)
Hi welcome to learn Linux.
This is the Linux Server 5
```

LINUX SERVER 6

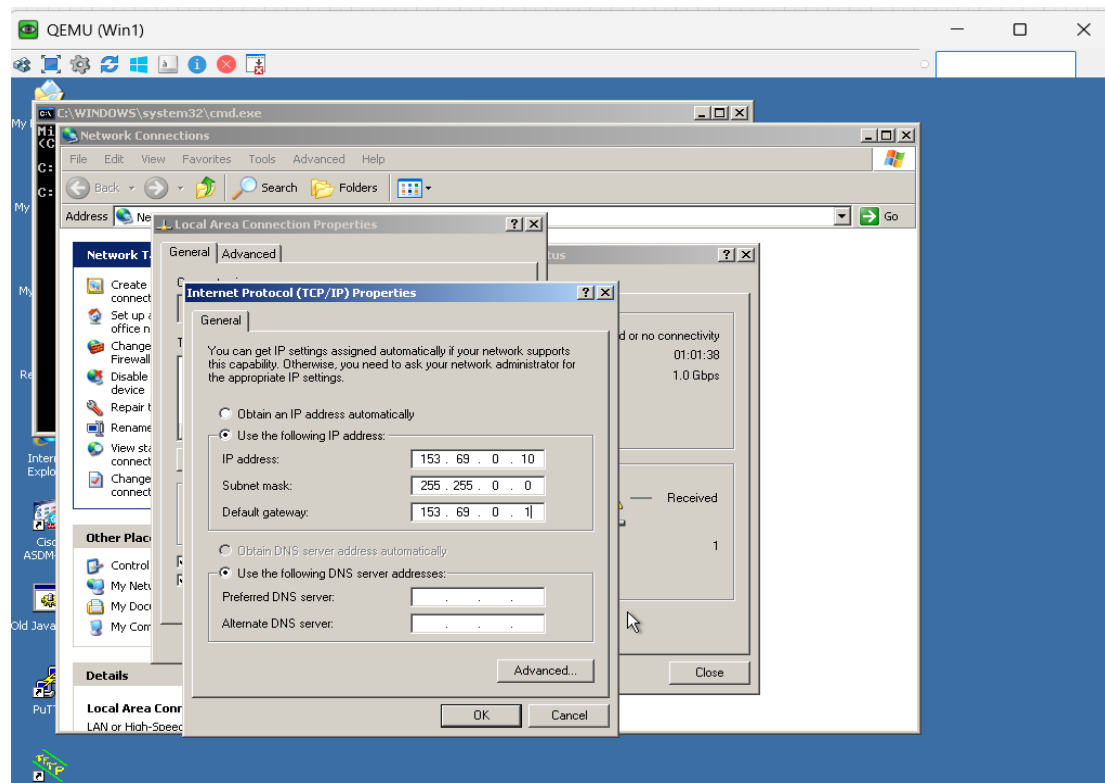
```
QEMU (Linux6)
Hi welcome to learn Linux.
This is the Linux Server 6

/home/htdocs/index.html [Modified] 1/2 50%
```

TASK 4: ASSIGNING IP ADDRESS TO PCs

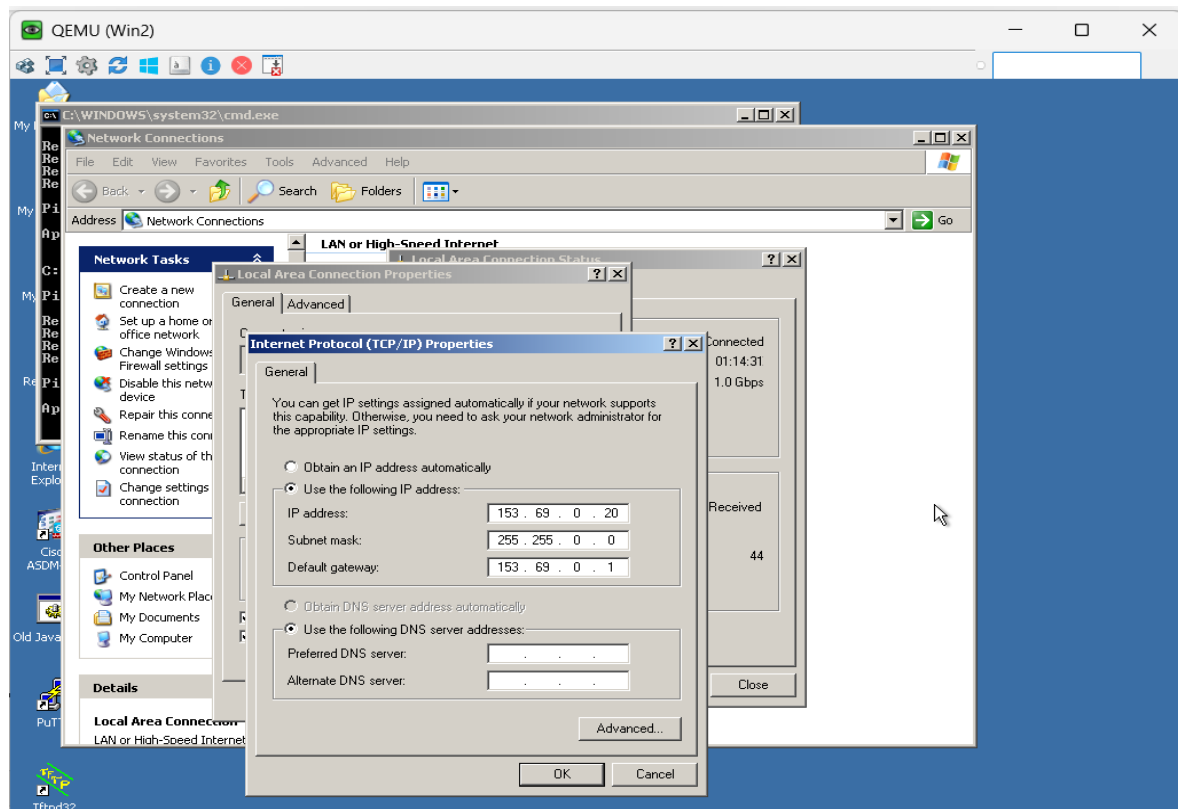
PC1 IP: 153.69.0.10

GATEWAY: 153.69.0.1



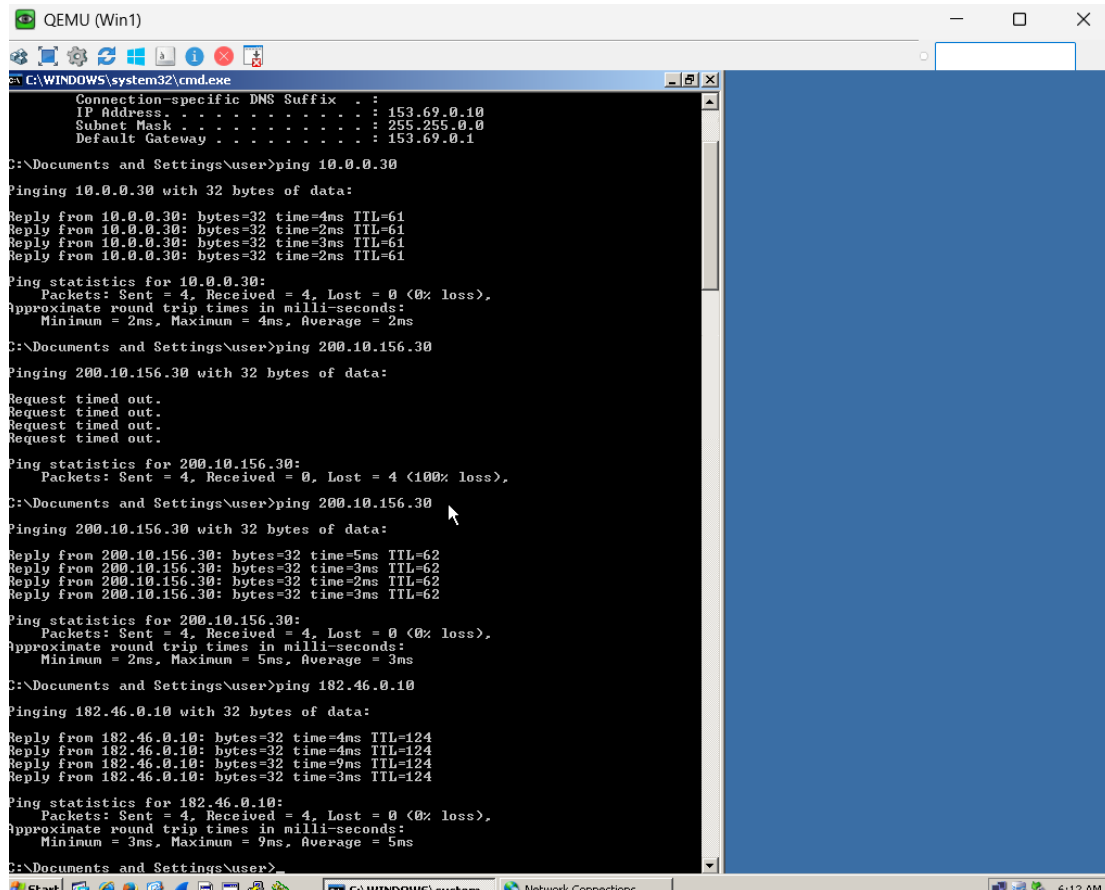
PC2 IP: 153.69.0.20

GATEWAY: 153.69.0.1



TASK 5: VERIFYING THE CONNECTION ESTABLISHMENTS USING ping

CLIENT PC 1



```
QEMU (Win1)
C:\WINDOWS\system32\cmd.exe
Connection-specific DNS Suffix  : 
IP Address. . . . . : 153.69.0.10
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 153.69.0.1

C:\Documents and Settings\User>ping 10.0.0.30

Pinging 10.0.0.30 with 32 bytes of data:

Reply from 10.0.0.30: bytes=32 time=4ms TTL=61
Reply from 10.0.0.30: bytes=32 time=2ms TTL=61
Reply from 10.0.0.30: bytes=32 time=3ms TTL=61
Reply from 10.0.0.30: bytes=32 time=2ms TTL=61

Ping statistics for 10.0.0.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\User>ping 200.10.156.30

Pinging 200.10.156.30 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 200.10.156.30:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\User>ping 200.10.156.30

Pinging 200.10.156.30 with 32 bytes of data:

Reply from 200.10.156.30: bytes=32 time=5ms TTL=62
Reply from 200.10.156.30: bytes=32 time=3ms TTL=62
Reply from 200.10.156.30: bytes=32 time=2ms TTL=62
Reply from 200.10.156.30: bytes=32 time=3ms TTL=62

Ping statistics for 200.10.156.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\Documents and Settings\User>ping 182.46.0.10

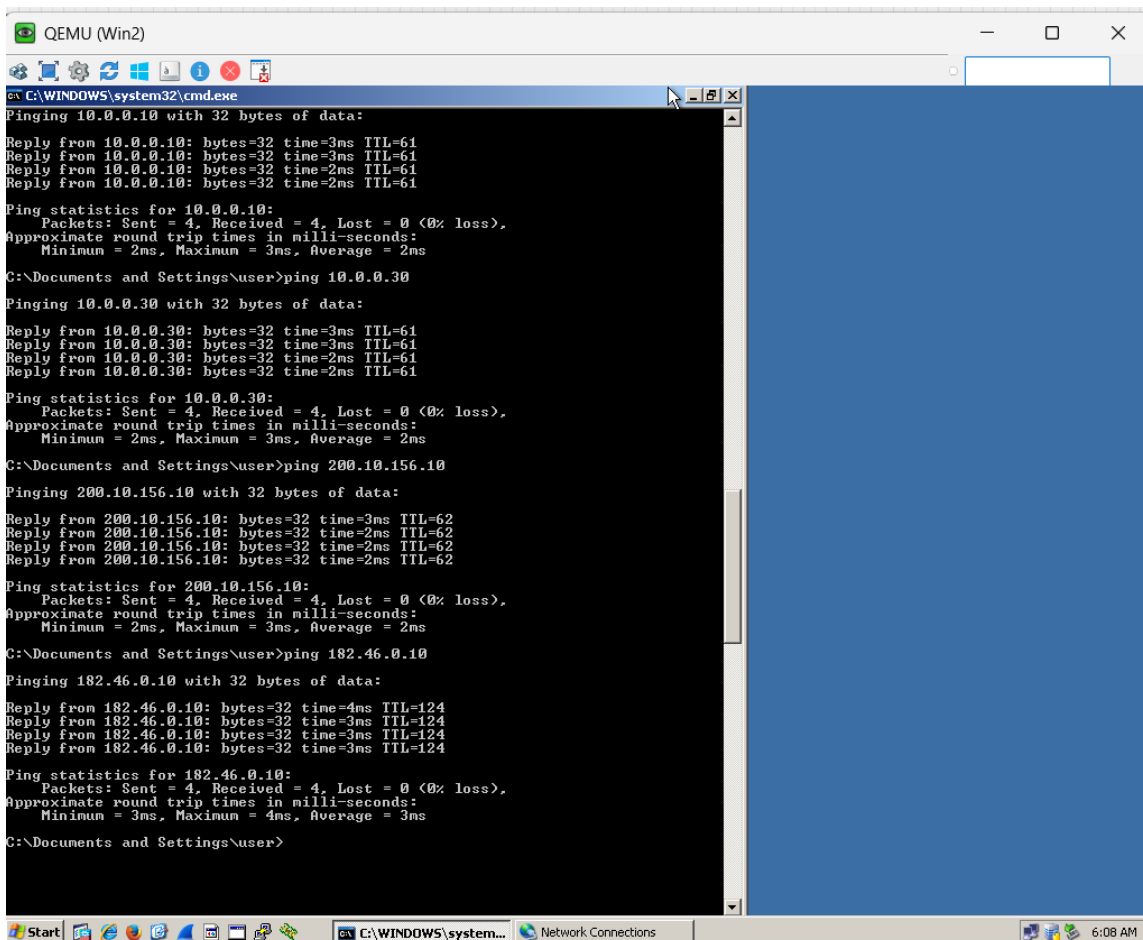
Pinging 182.46.0.10 with 32 bytes of data:

Reply from 182.46.0.10: bytes=32 time=4ms TTL=124
Reply from 182.46.0.10: bytes=32 time=4ms TTL=124
Reply from 182.46.0.10: bytes=32 time=3ms TTL=124
Reply from 182.46.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 182.46.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 5ms, Average = 5ms

C:\Documents and Settings\User>
```

CLIENT PC 2



```
QEMU (Win2)
C:\WINDOWS\system32\cmd.exe

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=61
Reply from 10.0.0.10: bytes=32 time=3ms TTL=61
Reply from 10.0.0.10: bytes=32 time=2ms TTL=61
Reply from 10.0.0.10: bytes=32 time=2ms TTL=61

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 10.0.0.30

Pinging 10.0.0.30 with 32 bytes of data:

Reply from 10.0.0.30: bytes=32 time=3ms TTL=61
Reply from 10.0.0.30: bytes=32 time=3ms TTL=61
Reply from 10.0.0.30: bytes=32 time=2ms TTL=61
Reply from 10.0.0.30: bytes=32 time=2ms TTL=61

Ping statistics for 10.0.0.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 200.10.156.10

Pinging 200.10.156.10 with 32 bytes of data:

Reply from 200.10.156.10: bytes=32 time=3ms TTL=62
Reply from 200.10.156.10: bytes=32 time=2ms TTL=62
Reply from 200.10.156.10: bytes=32 time=2ms TTL=62
Reply from 200.10.156.10: bytes=32 time=2ms TTL=62

Ping statistics for 200.10.156.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 182.46.0.10

Pinging 182.46.0.10 with 32 bytes of data:

Reply from 182.46.0.10: bytes=32 time=4ms TTL=124
Reply from 182.46.0.10: bytes=32 time=3ms TTL=124
Reply from 182.46.0.10: bytes=32 time=3ms TTL=124
Reply from 182.46.0.10: bytes=32 time=3ms TTL=124

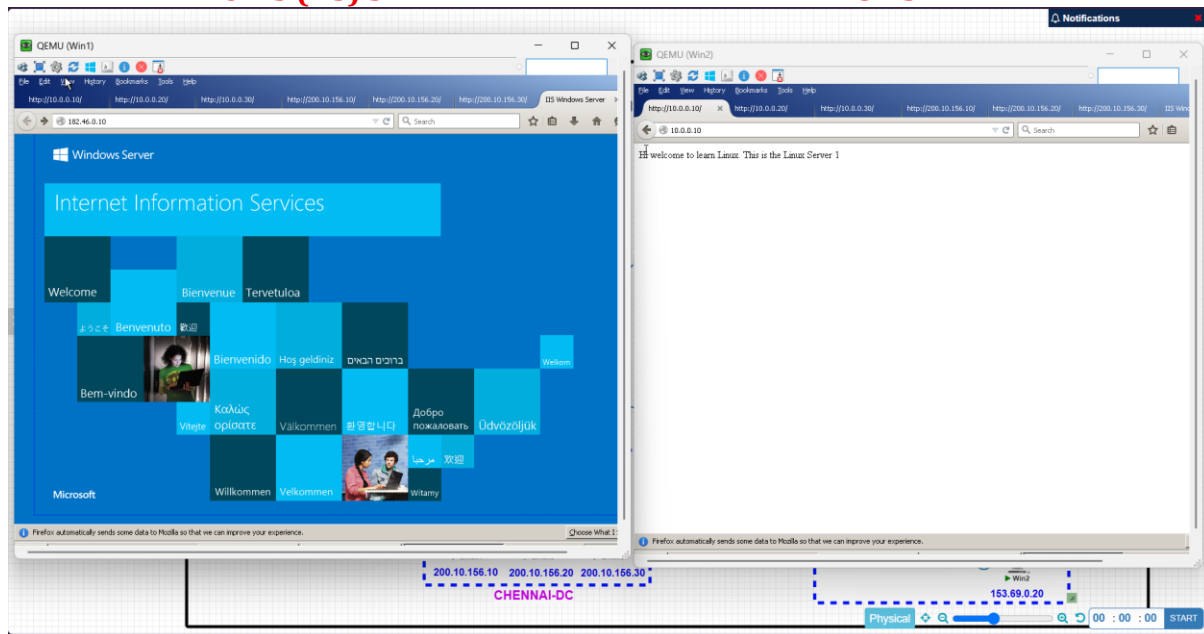
Ping statistics for 182.46.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\User>
```


TASK 6: VERIFYING THE WEB SERVERS FROM CLIENT PCs

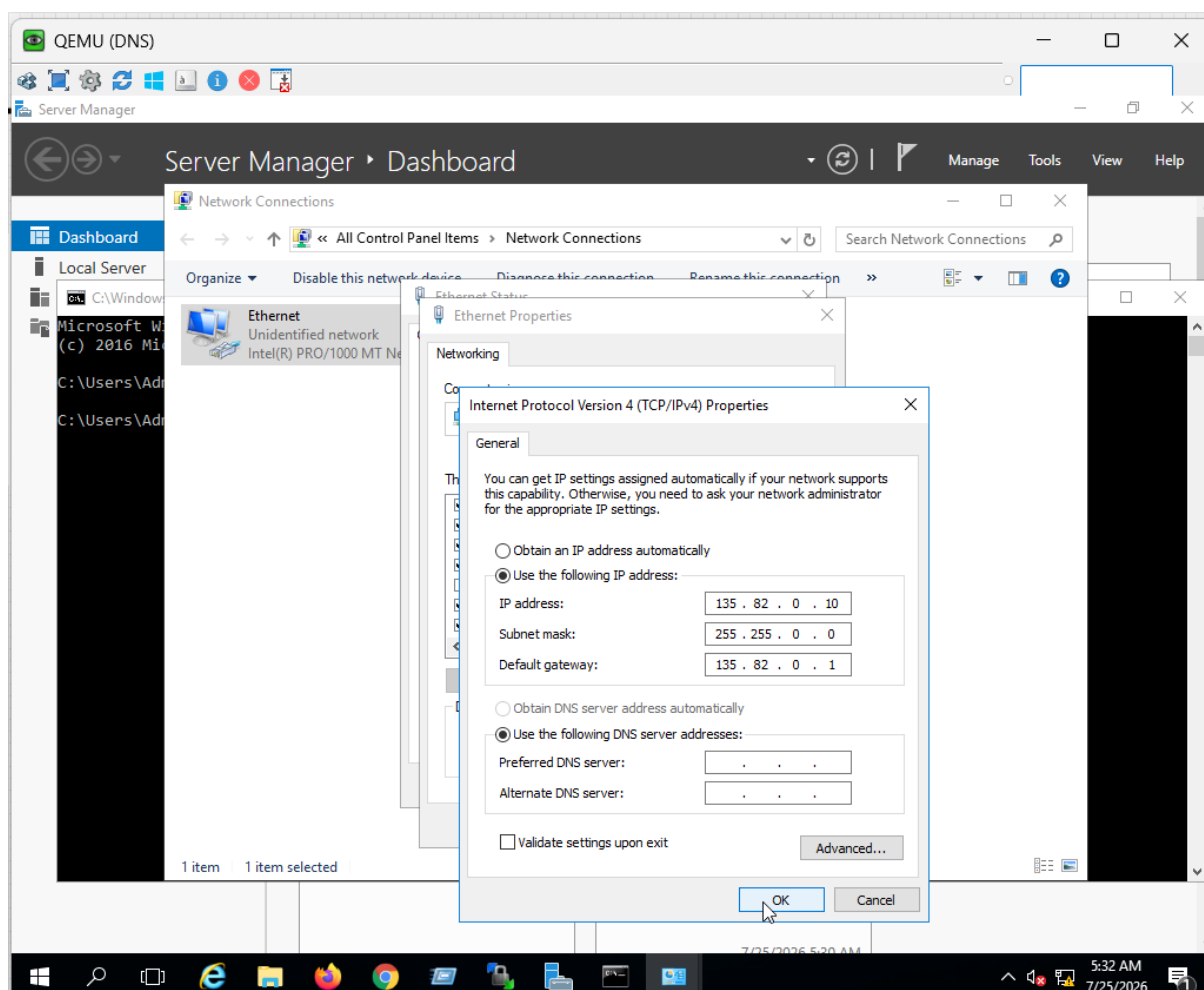
WINDOWS (IIS) SERVER

LINUX SERVER

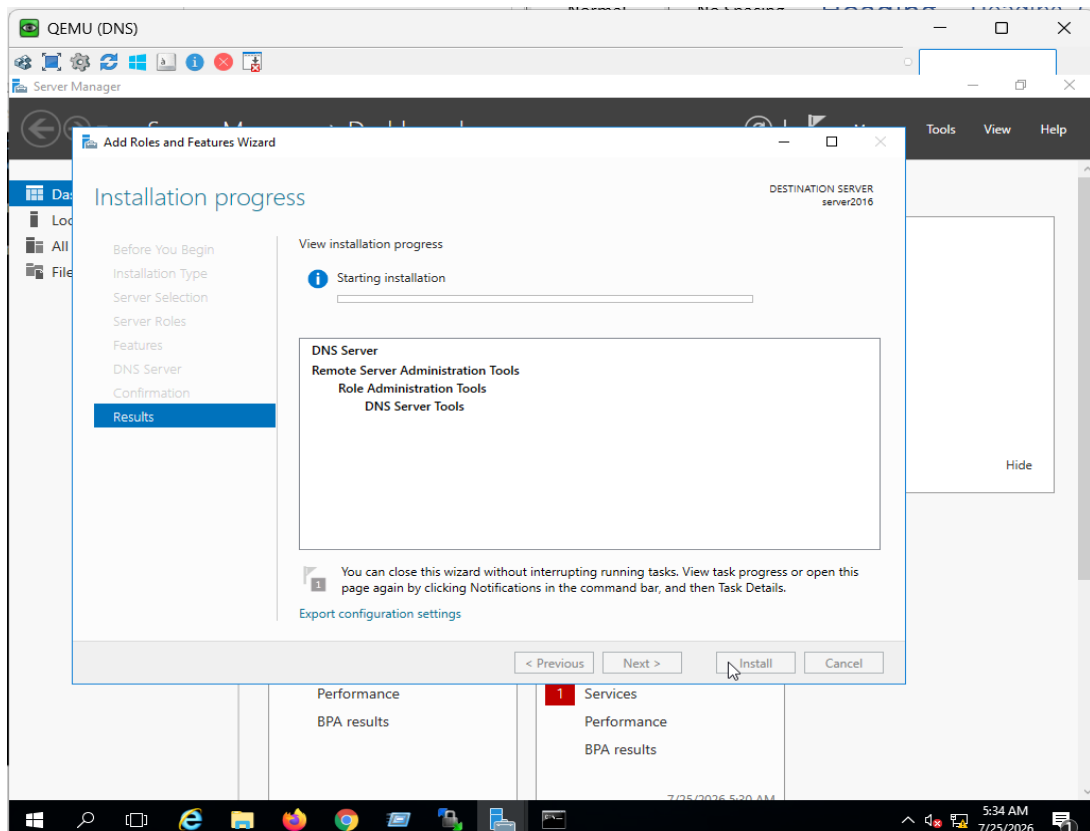


TASK 7: CONFIGURING THE DNS SERVER

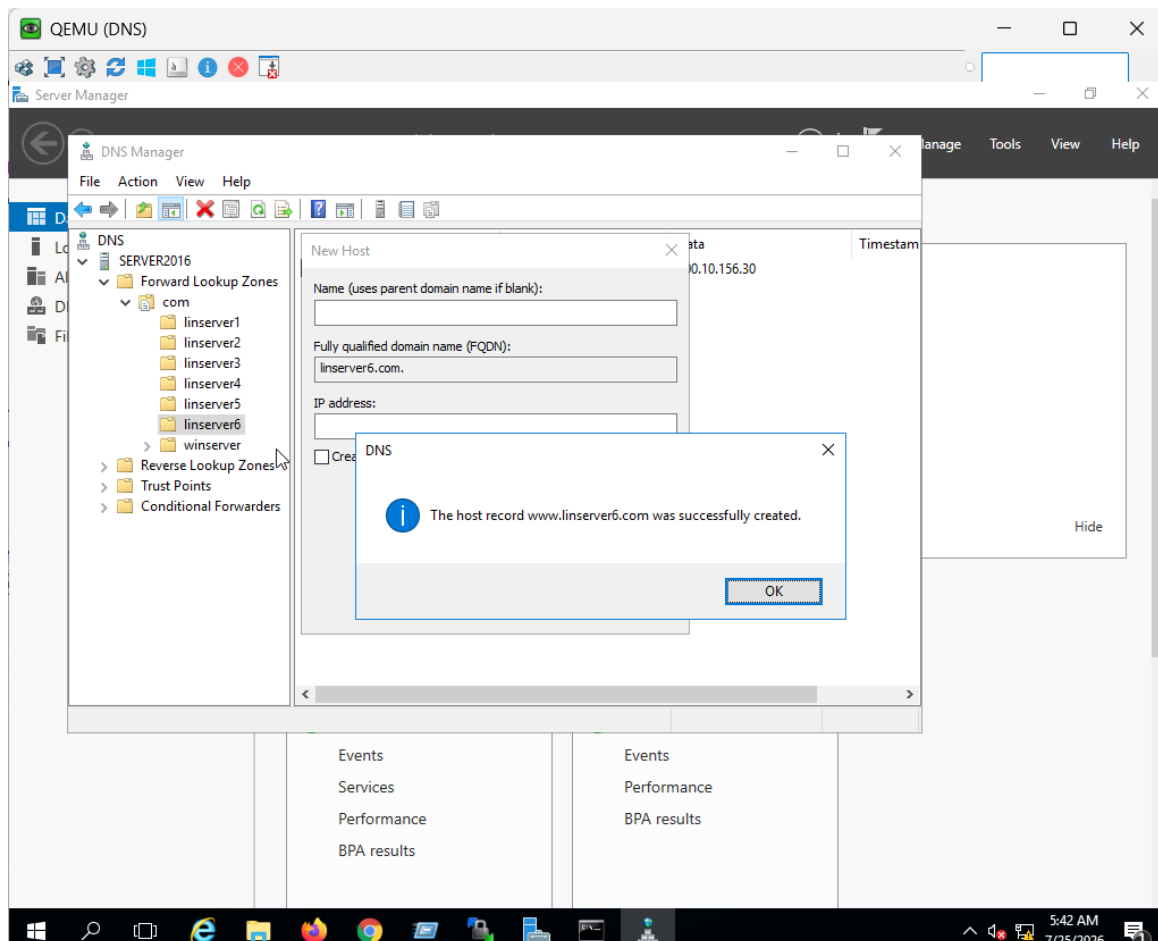
IP ASSIGNMENT



INSTALLING DNS SERVER



ADDING RECORDS TO THE DNS



TASK 8: VERIFYING THE DNS FROM THE CLIENT PCs

LINUX SERVER

WINDOWS (IIS) SERVER

